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MATHEMATICAL ANALYSIS / ACHIEVEMENT SETS / FRACTAL GEOMETRY

# AN ACHIEVABLE CANTORVAL WITH ZERO-DIMENSIONAL BOUNDARY

A factorial mixed-radix construction and a candidate affirmative solution to Problem 23(ii)

$\dim_{\mathrm{H}}(\partial E) = 0$

THE $n$TH BLOCK

$$B_n = \frac{1}{2^n(n+1)!}(3, \underbrace{2, \dots, 2}_{n})$$

MIXED-RADIX MECHANISM

$$b_n = 2n+2, \qquad |D_n| = b_n, \qquad D_n \pmod{b_n} = \{0, 1, \dots, b_n - 1\}$$

FORMULA-DERIVED LEVEL-0 THROUGH LEVEL-6 COVERS OF THE ACHIEVEMENT-SET BOUNDARY

FACTORIAL SCALE COLLAPSE

$$N_N \le 2 \cdot 3^N, \qquad \delta_N \le \frac{2}{3 \cdot 2^N(N+1)!}$$

Metriq PRISM Laboratory
A research laboratory of Metriq Foundation, Inc.

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# An Achievable Cantorval with Zero-Dimensional Boundary

## A factorial mixed-radix construction and a candidate affirmative solution to Problem 23(ii)

**Metriq PRISM Laboratory**
Program for Research in Intelligent Systems and Methods
A research laboratory of Metriq Foundation, Inc. • Florida, United States • metriq.org/prism
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**Research status**

This manuscript presents a candidate resolution of a published open problem. It has not undergone independent peer review. A December 2025 survey lists as Problem 23(ii) the existence of an achievable Cantorval whose boundary has Hausdorff dimension zero [2]. The question originates in work of Pratsiovytyi and Karvatskyi [6], and a March 2026 paper again states that integer-dimensional Cantorval boundaries remain open [4]. A targeted literature search completed on 16 July 2026 did not identify a later resolution or the construction below; that search does not establish publication priority. Independent specialist verification is required before the result should be treated as established.

**Abstract**

For a summable positive sequence, its achievement set is the compact set of all subsums. We give a nonstationary mixed-radix construction intended to answer affirmatively the question whether an achievable Cantorval can have boundary of Hausdorff dimension zero. Put

$$b_n = 2n + 2, \qquad Q_0 = 1, \qquad Q_n = \prod_{k=1}^{n} b_k = 2^n(n+1)!,$$

and concatenate the blocks

$$B_n = \left( \frac{3}{Q_n}, \underbrace{\frac{2}{Q_n}, \dots, \frac{2}{Q_n}}_{n \text{ copies}} \right), \qquad n \geq 1.$$

The subset sums in block $n$ form the digit set

$$D_n = \{0, 2, 3, \dots, b_n - 1, b_n + 1\},$$

a complete residue system modulo $b_n$. A finite mixed-radix argument yields the explicit interval

$$[4\sqrt{e} - 6, 1] \subset E.$$

A recursive lower-branch decomposition produces infinitely many genuine gaps, so the Guthrie–Nymann classification implies that $E$ is a Cantorval. The same decomposition gives a two-state boundary recursion with at most three descendants per state at each level. After $N$ levels, $\partial E$ is covered by at most $2 \cdot 3^N$ intervals of diameter at most $2/(3Q_N)$. Since $Q_N = 2^N(N+1)!$, the $s$-dimensional cost of these covers tends to zero for every $s > 0$. Hence $\dim_{\mathrm{H}}(\partial E) = 0$.

**2020 Mathematics Subject Classification.** Primary 40A05; Secondary 28A80, 11B05, 28A78.
**Keywords.** Achievement set; set of subsums; Cantorval; Hausdorff dimension zero; boundary; mixed-radix expansion; factorial scale.

**Candidate main result.** There exists a summable positive sequence whose achievement set $E$ is a Cantorval and satisfies

$$\dim_{\mathrm{H}}(\partial E) = 0.$$

The explicit sequence is the concatenation of the blocks $B_n$ displayed above.

# 1 Introduction

Let $x = (x_k)_{k\geq 1}$ be a summable sequence of positive real numbers. Its *achievement set*, or *set of subsums*, is

$$\mathscr{E}(x) = \left\{ \sum_{k=1}^{\infty} \varepsilon_k x_k \;:\; \varepsilon_k \in \{0,1\} \right\}.$$

It is compact and perfect. The classification theorem of Guthrie and Nymann places every such set into one of three relevant topological classes: a finite union of closed intervals, a Cantor set, or a Cantorval [3]. Nymann and Sáenz later supplied a complete corrected proof of the theorem [5]. In this manuscript, a *Cantorval* means the third type: a regularly closed compact set with infinitely many gaps and Cantor boundary.

The fractal dimension of that boundary has only recently been studied systematically. Pratsiovytyi and Karvatskyi computed fractional dimensions for the Guthrie–Nymann example and related families, and explicitly asked whether the boundary can be an "anomalyfractal" of Hausdorff dimension zero [6]. The 2025 survey of Głąb and Prus-Wiśniowski records the question as follows [2, Problem 23]:

**Problem 23(ii)**

Does there exist an achievable Cantorval $E$ such that

$$\dim_{\mathrm{H}}(\mathrm{Fr}\, E) = 0?$$

Known stationary self-similar examples have positive fractional boundary dimension. For example, Karvatskyi's family has

$$\dim_{\mathrm{H}}(\partial K_\ell) = \frac{\log(2\ell+1)}{\log(2\ell+2)},$$

which is strictly between zero and one for each fixed parameter [4]. The construction below instead lets the radix increase with the scale. Boundary branching remains at most exponential, while geometric contraction is factorial. This mismatch is the mechanism that drives the dimension to zero.

The proof has three parts:

1. complete residue systems force an explicit central interval;
2. a separated two-branch recursion forces infinitely many gaps;
3. a three-descendant boundary recursion gives factorially shrinking covers.

# 2 The construction and its normalized tails

Table 1: The first five blocks. Numerators are shown before division by $Q_n$.

| $n$ | $b_n$ | $Q_n$ | block numerators | $D_n$ |
|---|---|---|---|---|
| 1 | 4 | 4 | $(3, 2)$ | $\{0, 2, 3, 5\}$ |
| 2 | 6 | 24 | $(3, 2, 2)$ | $\{0, 2, 3, 4, 5, 7\}$ |
| 3 | 8 | 192 | $(3, 2, 2, 2)$ | $\{0, 2, 3, 4, 5, 6, 7, 9\}$ |
| 4 | 10 | 1920 | $(3, 2, 2, 2, 2)$ | $\{0, 2, 3, \dots, 9, 11\}$ |
| 5 | 12 | 23040 | $(3, 2, 2, 2, 2, 2)$ | $\{0, 2, 3, \dots, 11, 13\}$ |

## 2.1 Blocks, bases, and digits

For $n \geq 1$, define

$$b_n = 2n + 2, \qquad Q_0 = 1, \qquad Q_n = \prod_{k=1}^{n} b_k = 2^n (n+1)!. \tag{1}$$

Let $x$ be obtained by concatenating the blocks

$$B_n = \left( \frac{3}{Q_n}, \underbrace{\frac{2}{Q_n}, \dots, \frac{2}{Q_n}}_{n \text{ copies}} \right). \tag{2}$$

Within a block, $3/Q_n > 2/Q_n$. Between consecutive blocks,

$$\frac{2}{Q_n} > \frac{3}{Q_{n+1}},$$

because $b_{n+1} \geq 4$. Thus $x$ is nonincreasing.

The total of block $n$ is $(2n+3)/Q_n = (b_n + 1)/Q_n$. The series converges, for example because

$$\frac{b_n + 1}{Q_n} < \frac{2b_n}{Q_n} = \frac{2}{Q_{n-1}},$$

and $Q_n$ grows factorially. We will obtain the exact total below.

Selecting $j$ of the $n$ equal terms, with or without the term $3/Q_n$, produces the integer digit set

$$D_n = \{0, 2, 4, \dots, 2n\} \cup \{3, 5, \dots, 2n+3\} = \{0, 2, 3, \dots, b_n - 1, b_n + 1\}. \tag{3}$$

Consequently,

$$E := \mathscr{E}(x) = \left\{ \sum_{n=1}^{\infty} \frac{d_n}{Q_n} \;:\; d_n \in D_n \right\}. \tag{4}$$

**Lemma 2.1** (Residues and symmetry). *For every $n$, the set $D_n$ has $b_n$ elements and contains exactly one representative of every residue class modulo $b_n$. Moreover,*

$$d \in D_n \quad \Longleftrightarrow \quad b_n + 1 - d \in D_n.$$

*Proof.* Modulo $b_n$, the digits are $0, 2, 3, \dots, b_n - 1, 1$. These are all residue classes exactly once. The reflection statement follows directly from the displayed form of $D_n$. □

## 2.2 Normalized tail sets

For $n \le k$, put

$$B_{n,k} = \prod_{j=n}^{k} b_j, \qquad B_{n,n-1} = 1,$$

and define

$$A_n = \sum_{k=n}^{\infty} \frac{1}{B_{n,k}}, \qquad E_n = \left\{ \sum_{k=n}^{\infty} \frac{d_k}{B_{n,k}} \;:\; d_k \in D_k \right\}. \tag{5}$$

Thus $E = E_1$. Let $S_n = \max E_n$. From the definitions,

$$A_n = \frac{1 + A_{n+1}}{b_n}, \qquad S_n = 1 + 2A_n, \qquad E_n = \bigcup_{d \in D_n} \frac{d + E_{n+1}}{b_n}. \tag{6}$$

Indeed,

$$\sum_{k=n}^{\infty} \frac{b_k + 1}{B_{n,k}} = \sum_{k=n}^{\infty} \frac{1}{B_{n,k-1}} + \sum_{k=n}^{\infty} \frac{1}{B_{n,k}} = 1 + 2A_n.$$

Since all $b_k \ge 4$,

$$0 < A_n \le \sum_{j=1}^{\infty} 4^{-j} = \frac{1}{3}, \qquad S_n \le \frac{5}{3} < 2. \tag{7}$$

For the explicit bases, a change of index gives

$$A_1 = \sum_{n=1}^{\infty} \frac{1}{2^n (n+1)!} = 2\sqrt{e} - 3, \qquad S_1 = 1 + 2A_1 = 4\sqrt{e} - 5. \tag{8}$$

Thus the total sum of the original positive series is $4\sqrt{e} - 5$.

# 3 A complete-residue interval theorem

We now prove that every normalized tail $E_n$ contains a central interval. The argument is finite at each stage and does not invoke a category theorem.

Fix $n \le N$. Write

$$P = B_{n,N}, \qquad w_k = B_{k+1,N} \quad (n \le k \le N),$$

where $w_N = 1$, and define the integer prefix set

$$\mathscr{P}_{n,N} = \left\{ \sum_{k=n}^{N} d_k w_k \;:\; d_k \in D_k \right\}. \tag{9}$$

**Lemma 3.1** (Mixed-radix bijection). *Reduction modulo $P$ maps the digit words in (9) bijectively onto $\mathbb{Z}/P\mathbb{Z}$.*

*Proof.* Suppose two prefix sums are congruent modulo $P$. Reducing first modulo $b_N$ leaves $d_N \equiv d'_N \pmod{b_N}$. By theorem 2.1, the representatives in $D_N$ are unique, so $d_N = d'_N$. Subtract and divide by $b_N$. Repeating from right to left gives equality of every digit. Thus the residue map is injective. The number of words is

$$\prod_{k=n}^{N} |D_k| = \prod_{k=n}^{N} b_k = P,$$

so the map is bijective. □

Set

$$C_{n,N} = 2\sum_{k=n}^{N} w_k. \tag{10}$$

The canonical mixed-radix digits $b_k - 1$ have total $P - 1$. Since the maximum allowed digit is $b_k + 1$, the largest member of $\mathscr{P}_{n,N}$ is

$$P - 1 + C_{n,N}. \tag{11}$$

Moreover,

$$\frac{C_{n,N}}{P} = 2\sum_{k=n}^{N} \frac{1}{B_{n,k}} \le 2A_n < 1.$$

**Lemma 3.2** (A consecutive prefix block)**.** *Every integer in* $[C_{n,N}, P-1]$ *belongs to* $\mathscr{P}_{n,N}$*.*

*Proof.* Take $r \in [C_{n,N}, P-1]$. By theorem 3.1, its unique prefix representative has the form $r + mP$ with $m \ge 0$. If $m \ge 1$, then

$$r + mP \ge C_{n,N} + P > P - 1 + C_{n,N},$$

contradicting (11). Hence $m = 0$ and $r \in \mathscr{P}_{n,N}$. □

Define the compact outer approximation

$$K_{n,N} = \bigcup_{p \in \mathscr{P}_{n,N}} \left[\frac{p}{P}, \frac{p + S_{N+1}}{P}\right]. \tag{12}$$

The sets $K_{n,N}$ are nested, contain $E_n$, and satisfy

$$E_n = \bigcap_{N \ge n} K_{n,N}. \tag{13}$$

For the reverse inclusion, note that a point of $K_{n,N}$ is at distance at most $S_{N+1}/P$ from a finite prefix belonging to $E_n$. This bound tends to zero; since $E_n$ is closed, every point in the intersection belongs to $E_n$.

**Proposition 3.3** (Explicit central interval)**.** *For every* $n \ge 1$*,*

$$[2A_n, 1] \subset E_n.$$

*In particular,*

$$\boxed{[4\sqrt{e} - 6, 1] \subset E.}$$

*Proof.* By theorem 3.2, the intervals in (12) indexed by

$$C_{n,N}, C_{n,N} + 1, \dots, P - 1$$

form one interval, because $S_{N+1} > 1$. Their union is

$$\left[2\sum_{k=n}^{N} \frac{1}{B_{n,k}}, 1 + \frac{2A_{N+1}}{P}\right].$$

Every $x \in [2A_n, 1]$ lies in this interval for every $N$: the left endpoint is at most $2A_n$, and the right endpoint is at least 1. Hence $x \in K_{n,N}$ for all $N$. Equation (13) gives $x \in E_n$. The explicit endpoint follows from (8). □

Figure 1 plots the exact finite-level outer approximations for the first five levels. The component counts $3, 9, 27, 81, 243$ are finite consistency checks of the recursive geometry developed next.

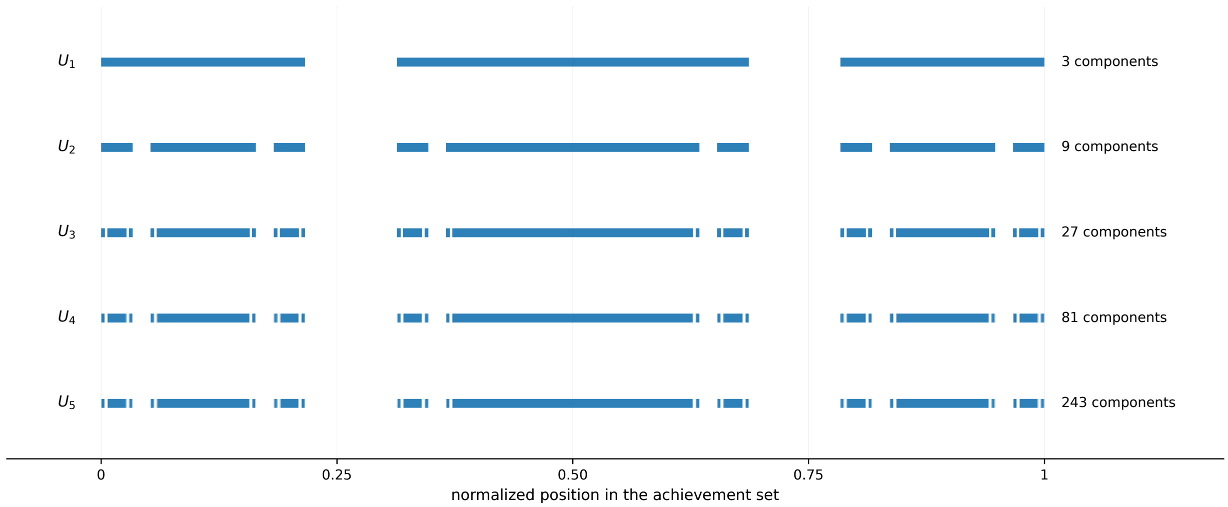


Figure 1: Finite-level outer approximations $U_N = K_{1,N}$, normalized by the total sum. The central overlap survives while the two outer regions split into three descendants per level. The plotted endpoints are computed from the exact mixed-radix prefixes and high-precision tails.

## 4 Separated outer branches and the Cantorval property

Put

$$L_n = 2A_n, \qquad E_n^- = E_n \cap [0, L_n]. \tag{14}$$

The interval in theorem 3.3 is $[L_n, 1]$.

**Proposition 4.1** (Lower-branch recursion). *For every $n$,*

$$E_n^- = \frac{1}{b_n} E_{n+1} \,\dot\cup\, \frac{2 + E_{n+1}^-}{b_n}. \tag{15}$$

*The two compact pieces are separated by the open gap*

$$\left( \frac{S_{n+1}}{b_n}, \frac{2}{b_n} \right). \tag{16}$$

*Proof.* By (6),

$$L_n = \frac{2 + 2A_{n+1}}{b_n}.$$

The digit-0 copy has maximum $S_{n+1}/b_n = (1 + 2A_{n+1})/b_n < L_n$, so it lies entirely in $E_n^-$. A point in the digit-2 copy lies below $L_n$ exactly when its tail coordinate is at most $2A_{n+1}$, giving the second term in (15). Every remaining digit is at least 3, and

$$\frac{3}{b_n} > \frac{2 + 2A_{n+1}}{b_n} = L_n$$

because $A_{n+1} < 1/2$. Finally, $S_{n+1} < 2$ by (7), proving the gap and the disjointness. □

The digit symmetry in theorem 2.1 induces

$$h_n(x) = S_n - x, \qquad h_n(E_n) = E_n. \tag{17}$$

Since $h_n(L_n) = 1$, the upper outer part

$$E_n^+ := E_n \cap [1, S_n]$$

is the reflected copy $h_n(E_n^-)$. Hence

$$E_n = E_n^- \cup [L_n, 1] \cup E_n^+. \tag{18}$$

Iterating the second branch in (15) produces global gaps. For $m \geq n$, set

$$p_{n,m} = 2\sum_{k=n}^{m} \frac{1}{B_{n,k}}, \qquad p_{n,n-1} = 0.$$

Then

$$G_{n,m} = \left(p_{n,m-1} + \frac{S_{m+1}}{B_{n,m}}, p_{n,m}\right) \subset \mathbb{R} \setminus E_n. \tag{19}$$

Indeed, (19) is the gap (16) at level $m$, transported through the digit-2 branch at all preceding levels. Its length is

$$|G_{n,m}| = \frac{2 - S_{m+1}}{B_{n,m}} > 0. \tag{20}$$

The gaps are pairwise disjoint: $G_{n,m}$ ends at $p_{n,m}$, while $G_{n,m+1}$ begins strictly to the right of that point. Their upper endpoints satisfy

$$p_{n,m} \uparrow 2A_n = L_n.$$

**Corollary 4.2.** *The achievement set $E$ is a Cantorval.*

*Proof.* By theorem 3.3, $E$ has nonempty interior and is therefore not a Cantor set. By (19) with $n = 1$, it has infinitely many gaps and therefore is not a finite union of intervals. The Guthrie–Nymann classification theorem, with the corrected proof of Nymann and Sáenz, leaves the Cantorval alternative [3, 5]. □

# 5 A finite-state recursion for the boundary

Let

$$F_n = \partial E_n, \qquad F_n^- = F_n \cap [0, L_n], \qquad F_n^+ = F_n \cap [1, S_n].$$

Since $(L_n, 1) \subset \operatorname{int} E_n$, one has

$$F_n = F_n^- \cup F_n^+. \tag{21}$$

The gaps $G_{n,m}$ accumulate at $L_n$, so $L_n \in F_n$. By symmetry, $1 = h_n(L_n) \in F_n$ as well.

For $d \in D_n$, write

$$f_{n,d}(x) = \frac{d + x}{b_n}.$$

**Proposition 5.1** (Boundary recursion)**.** *For every $n$,*

$$F_n^- = f_{n,0}(F_{n+1}) \cup f_{n,2}(F_{n+1}^-) \tag{22}$$

*and therefore*

$$F_n^- = f_{n,0}(F_{n+1}^-) \cup f_{n,0}(F_{n+1}^+) \cup f_{n,2}(F_{n+1}^-). \tag{23}$$

*By reflection, $F_n^+$ is likewise a union of three affine images of $F_{n+1}^-$ and $F_{n+1}^+$, each with contraction ratio $1/b_n$.*

*Proof.* The lower set has the separated decomposition in (15). On the first component $f_{n,0}(E_{n+1})$, local interior and boundary are carried exactly by the affine homeomorphism $f_{n,0}$; the component is isolated from the rest of $E_n$ by (16). This contributes $f_{n,0}(F_{n+1})$.

On the second component, every point below $L_n$ has a neighborhood in which $E_n$ is precisely the affine image of $E_{n+1}^-$. Thus its boundary points are $f_{n,2}(F_{n+1}^-)$ away from the endpoint. The endpoint itself is $L_n = f_{n,2}(L_{n+1})$; both $L_n$ and $L_{n+1}$ are boundary points by the accumulating-gap argument above. This proves (22). Equation (23) follows from (21), and the statement for $F_n^+$ follows by conjugating with the symmetry maps $h_n$ and $h_{n+1}$. □

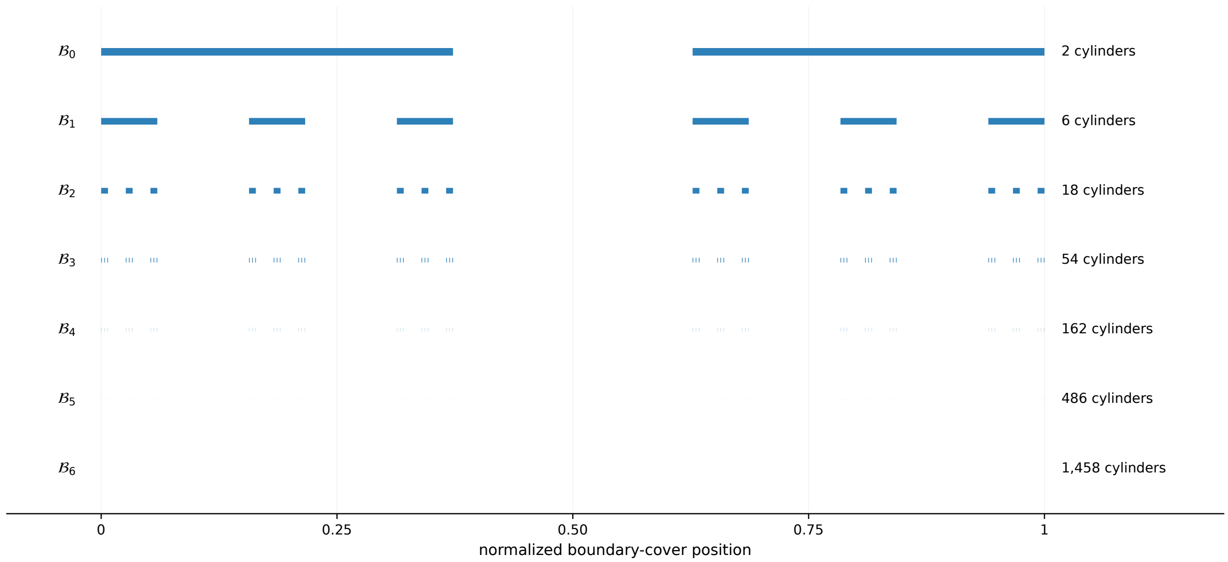


Figure 2: The first seven finite-state covers $\mathscr{B}_N$ of the boundary, normalized by the total sum. The number of cylinders is $2 \cdot 3^N$, but their diameter is bounded by $2/(3Q_N)$ with $Q_N = 2^N(N+1)!$. The visual is generated directly from the affine state transitions in theorem 5.1.

Starting with the two states $F_1^-$ and $F_1^+$, each application of theorem 5.1 creates at most three descendant states. Consequently, after $N$ levels, $F_1$ is covered by at most

$$2 \cdot 3^N \tag{24}$$

affine images of $F_{N+1}^-$ or $F_{N+1}^+$, each contracted by $Q_N^{-1}$. The two tail states have diameter at most

$$\max\{L_{N+1}, S_{N+1} - 1\} = 2A_{N+1} \leq \frac{2}{3}.$$

Therefore every level-$N$ boundary cylinder has diameter at most

$$\delta_N = \frac{2}{3Q_N}. \tag{25}$$

## 6 Zero Hausdorff dimension

**Theorem 6.1** (Main theorem)**.** *Let $x$ be the positive summable sequence obtained by concatenating the blocks (2), and let $E = \mathscr{E}(x)$. Then $E$ is a Cantorval and*

$$\boxed{\dim_{\mathrm{H}}(\partial E) = 0.}$$

*Proof.* The Cantorval assertion is theorem 4.2. Fix $s > 0$. The cover in (24)–(25) gives

$$\mathscr{H}^s_{\delta_N}(\partial E) \leq 2 \cdot 3^N \left(\frac{2}{3Q_N}\right)^s. \tag{26}$$

Since $Q_N = 2^N(N+1)!$, the logarithm of the right-hand side is

$$O(1) + N\log 3 - sN\log 2 - s\log((N+1)!).$$

The final term dominates the linear terms. For instance,

$$(N+1)! \geq \left(\frac{N}{2}\right)^{N/2}$$

for all sufficiently large $N$, so the expression tends to $-\infty$. Thus the right-hand side of (26) tends to zero. Because $\delta_N \to 0$,

$$\mathscr{H}^s(\partial E) = 0$$

for every $s > 0$. The boundary is nonempty, and hence its Hausdorff dimension is exactly zero. □

**Corollary 6.2.** *Subject to independent verification and the literature-status caveat stated above, theorem 6.1 gives an affirmative answer to Problem 23(ii) of Głąb and Prus-Wiśniowski [2].*

The result is not saying that the boundary is finite or countable. Since $E$ is a Cantorval, its boundary is a Cantor set: compact, perfect, totally disconnected, and uncountable. The conclusion is that this Cantor set is metrically as thin as a nonempty set can be in Hausdorff dimension.

## 7 A variable-base family

The proof is not tied to $b_n = 2n+2$. It applies to a broader class and isolates the branching-versus-contraction principle.

**Theorem 7.1** (Even-base family)**.** *Let $(b_n)$ be any sequence of even integers with $b_n \geq 4$, and put $Q_n = \prod_{k=1}^{n} b_k$. Form the positive sequence by concatenating*

$$\left(\frac{3}{Q_n}, \underbrace{\frac{2}{Q_n}, \dots, \frac{2}{Q_n}}_{(b_n-2)/2 \text{ copies}}\right).$$

*Its achievement set $E$ is a Cantorval. Moreover,*

$$\dim_{\mathrm{H}}(\partial E) \leq \liminf_{N\to\infty} \frac{N\log 3}{\log Q_N}. \tag{27}$$

*In particular, if $\log Q_N/N \to \infty$, then $\dim_{\mathrm{H}}(\partial E) = 0$.*

*Proof.* The block digit set is again

$$\{0, 2, 4, \dots, b_n - 2\} \cup \{3, 5, \dots, b_n + 1\} = \{0, 2, 3, \dots, b_n - 1, b_n + 1\}.$$

Thus all complete-residue, symmetry, central-interval, lower-branch, gap, and boundary-recursion arguments above remain valid. In particular, the boundary has covers by at most $2 \cdot 3^N$ intervals of diameter at most $2/(3Q_N)$.

Let the liminf in (27) be $\alpha$. For any $s > \alpha$, choose a subsequence along which $N\log 3 \leq (s-\varepsilon)\log Q_N$ for some $\varepsilon > 0$. Along this subsequence, the $s$-cost of the boundary cover is at most a constant times $Q_N^{-\varepsilon}$ and therefore tends to zero. Hence $\dim_{\mathrm{H}}(\partial E) \leq s$. Letting $s \downarrow \alpha$ proves (27). The final statement is immediate. □

For constant base $b$, the estimate becomes $\dim_{\mathrm{H}}(\partial E) \leq \log_b 3$, consistent with the familiar three-branch boundary geometry. The present example makes $b_n$ grow, so the numerator in (27) remains linear while $\log Q_N$ grows like $N\log N$.

## 8 Verification priorities and limitations

All infinite assertions in this manuscript are argued symbolically. The supplementary script checks, with exact integer arithmetic or high-precision arithmetic where the closed form involves $e$, the following finite consequences:

1. the complete-residue and digit-symmetry properties for initial blocks;
2. the mixed-radix bijection and consecutive-prefix interval through level five;
3. the exact closed forms $A_1 = 2\sqrt{e} - 3$ and $S_1 = 4\sqrt{e} - 5$;
4. the component counts $3^N$ for the first five outer approximations;
5. the boundary-state count $2 \cdot 3^N$ and the decay diagnostics for the Hausdorff covers.

These computations are consistency checks, not evidence for the infinite claims.

The result has not been checked by an independent mathematician. The most important points for external review are:

1. the passage from complete residues to the consecutive prefix block in theorem 3.2;
2. the nested outer-approximation identity (13);
3. the global nature of the transported gaps (19);
4. the exact boundary recursion in theorem 5.1, especially the central endpoints;
5. the applicability of the corrected Guthrie–Nymann classification theorem;
6. possible prior art equivalent to the even-base family in theorem 7.1.

## Institutional disclaimer and computational research declaration

This manuscript is an exploratory research preprint issued by the Metriq PRISM Laboratory, an interdisciplinary research laboratory of Metriq Foundation, Inc. PRISM—the Program for Research in Intelligent Systems and Methods—conducts research involving computational, mathematical, scientific, and machine-assisted methods.

The work was developed as part of PRISM's experimental research program in computationally assisted mathematics. Metriq Foundation, acting through the PRISM Laboratory, directed the research process, evaluated the resulting argument, determined the form and scope of the manuscript, and accepts responsibility for its publication as a candidate result.

Generative artificial intelligence and other computational tools were used as research instruments to support construction search, symbolic and numerical analysis, literature review, proof development, verification, and manuscript preparation. Their use does not constitute independent validation of the results presented.

This manuscript has not yet undergone independent peer review and should not be regarded as an established resolution of the stated problem unless and until its arguments have been examined and confirmed by qualified mathematicians. Independent verification, criticism, replication, and identification of errors are expressly invited.

### Data, code, and reproducibility

No empirical dataset was used. The source package accompanying this preprint contains the LuaLaTeX manuscript, exact-arithmetic verification script, figure-generation script, generated figures, and a recorded verification output. The cover artwork is generated from the actual finite-state boundary covers and the exact central interval of the construction; it is not a stock image.

## Conflicts of interest

Metriq Foundation, Inc. declares no financial conflict of interest concerning the mathematical result presented in this preprint.

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